Nested Classes & Types of Classes

* Nested Classes
  - Static Nested
  - Inner class
  - local Class
  - Anonymous class

```
class A {
    class B {
        //code
    }
}
```

* why do we Need Nested Class?
  1. logical grouping
  2. Better access to outer class

```
class Outer {
    Private int x;
    class Inner {

    }
}
```

* Static Nested Class

```
class Outer {
    static class Inner {

    }
}
```

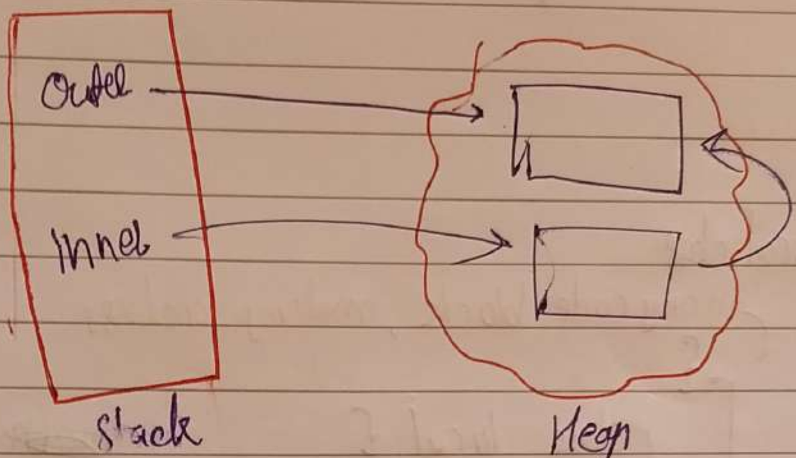
Key properties :
1. Does not need an instance of Outer class
2. class can be instantiated like Normal class
3. can access only static member of outer class
4. can access non static member by having a reference of outer class
5. It is just like normal class & can do anything an outer class does

Inner class

```
class Outer {
    class Inner {

    }
}
```

```
Outer outer = new Outer();
Outer.Inner inner = outer.new Inner();
```

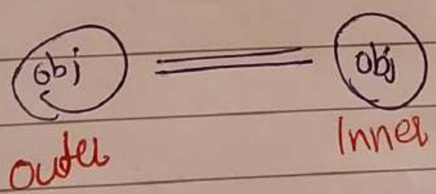



```
class Outer {
    class Inner{

    }
}
```

static member

non static " "

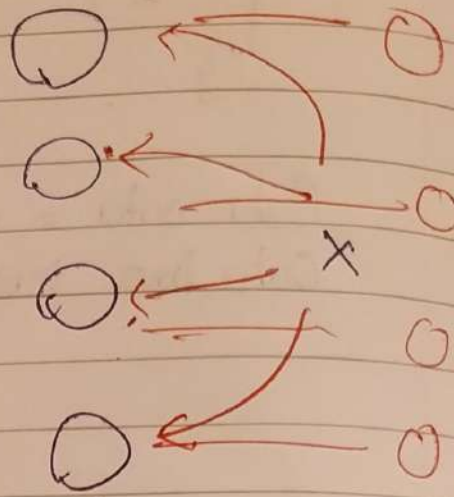



```
class Outer {
    int x = 10;
    class Inner{
      int x = 20;
    } sout(20); // 20
    } Sout(Outer.this.x); //10
    }
}
```

In old Java

- Inner class → Static members (not allowed) (X)

```
class A {
    class B {
        static int x;
    }
}
```



A

*local class

in any code block, making a class

```
{
    class local {

    }
}
```

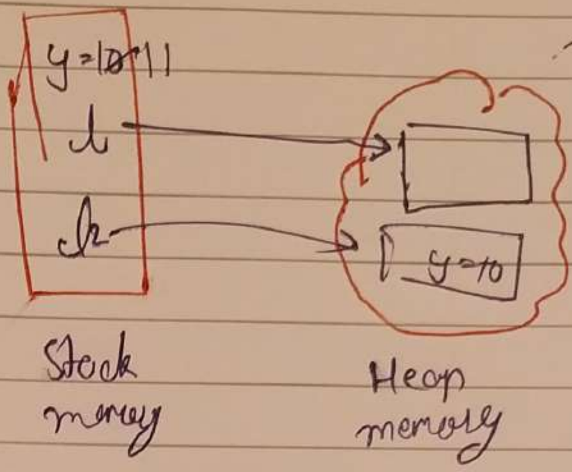
eg:

```
void fun () {
    class local {

    }
}
```

Effectively final rule in local class

```
class Outer {
    Object greet() {
        int y = 10;   y++;
        class Local {
            void fun() {
                sout(y);
            }
        }
        Local l = new Local();
        l.fun();
        return l;
    }
    void fun() {
        Object l2 = greet();
    }
}
```



* Anonymous Class
  - a class without name